

Daily Review



1 What is a system?

Parts working together
to do a job

2 Automatic doors: what is the input?

Sensing a person

3 Which energy does a marble have at the top of the run?

Gravitational potential

4 Write down the formula for power.

$P = E / t$ $W = J / s$

5 Name a real project from Booklet 1.

e.g. Edinburgh Trams

Marble Run — Class Feedback

What was good

- You explained clearly how your structure worked.
- You were honest about what went wrong.

What could be better

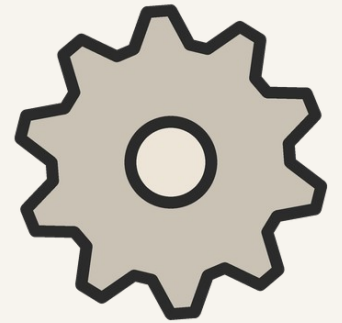
- Explain why things happened, in more detail.
e.g. “The marble slowed down because energy was lost to friction.”

Most of you showed respect to your team, and owned your own mistakes. I respect that.

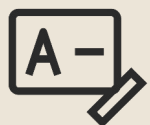
Engineers and What They Do

Who builds a tram — one engineer or four?

S3 Engineering Science



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.



Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

Who builds a tram, or a bridge — one engineer, or four?



Edinburgh Trams



The Queensferry Crossing

PART 1

Engineers and What They Do

Name them, describe what they do, explain how they work.

By the end of this part...



Rate each one 1-3 in your jotter: 1 not yet, 2 nearly, 3 got it.

- 1 I can name the 4 main types of engineering
- 2 I can describe what different engineers do
- 3 I can explain how engineers design, build, test and improve
- 4 I can explain my ideas clearly using what + why

SECTION A

Four Types of Engineer

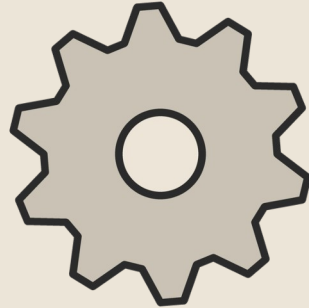
Who designs the tunnel, and who designs the paint?

The Four Types of Engineer



Electrical

electricity,
wiring, power
and control



Mechanical

moving parts,
machines and
forces



Civil

buildings,
roads,
bridges and
tunnels



Chemical

materials and
chemical
processes

One Project, Many Engineers – Edinburgh Trams

Electrical

power, lighting and displays

Mechanical

wheels, brakes and doors

Civil

tracks, stops and bridges

Chemical

paint and coatings that resist rust



Show Me — Which Engineer?



1

Which engineer designs a tunnel under a river?

Civil

2

...works on the gears of a wind turbine?

Mechanical

3

...designs the lighting for a football stadium?

Electrical

Show Me — Which Engineer?



4

...develops a paint that stops steel rusting?

Chemical

5

Name all four main types of engineering.

Electrical, mechanical,
civil, chemical

Show Me — Which Engineer?

EXTRA PRACTICE



- 1 Which engineer improves how drinking water is cleaned?
- 2 ...designs the brakes on a bike?
- 3 ...plans a new railway line?

Chemical

Mechanical

Civil

Show Me — Which Engineer?

EXTRA PRACTICE



4

...designs the control system for traffic lights?

Electrical

5

Why does a big project need more than one type of engineer?

Each works on a different part of it

SECTION B

Describe What Engineers Do

“Builds walls” — why does that score nothing?

Worked Example – Duns Park Flood Scheme

Describe one task a civil engineer would carry out. (1 mark)

What

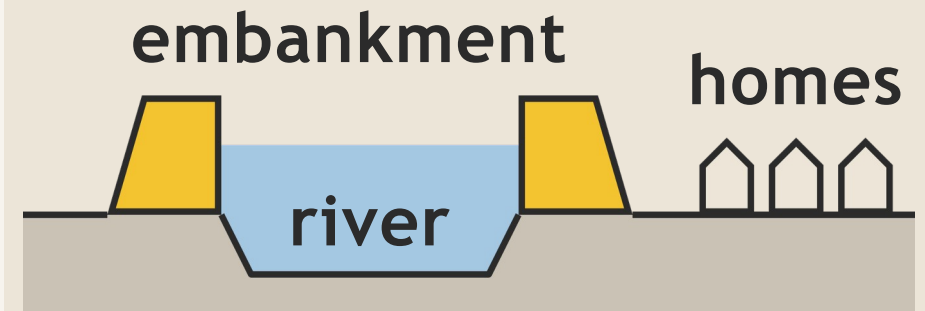
designs the flood embankments

Why

so nearby homes stay dry

Scores 0

“builds walls” – no detail, no why



Show Me — Queensferry Crossing



1 Civil engineer: one task, what + why

Designs the towers —
so the deck stays up

2 Electrical engineer: one task, what + why

Lights the deck — so
drivers see at night

3 Mechanical engineer: one task, what + why

Designs joints — so the
deck can expand

Show Me — Queensferry Crossing



4

Chemical engineer: one task, what + why

Chooses the paint — so
the steel won't rust

5

Which scores: A “fixes the lights” or
B “lights the deck so drivers see”?

B — it says what AND
why

Show Me — SpaceX Rocket

EXTRA PRACTICE



1

Electrical engineer: one task, what + why

Designs the controls —
so it can steer

2

Mechanical engineer: one task, what + why

Designs the engines —
so it can lift off

3

Chemical engineer: one task, what + why

Develops the fuel — **so it burns to give thrust**

Show Me — SpaceX Rocket

EXTRA PRACTICE



4

Improve: “an electrical engineer does the wiring”

5

Why must every engineer's part be done properly?

Designs the wiring — so **controls get power**

One failed part can stop the whole system

SECTION C

Design, Build, Test, Improve

Why did your marble run need more than one go?

The Design Cycle — Round Until It Works

Prototype

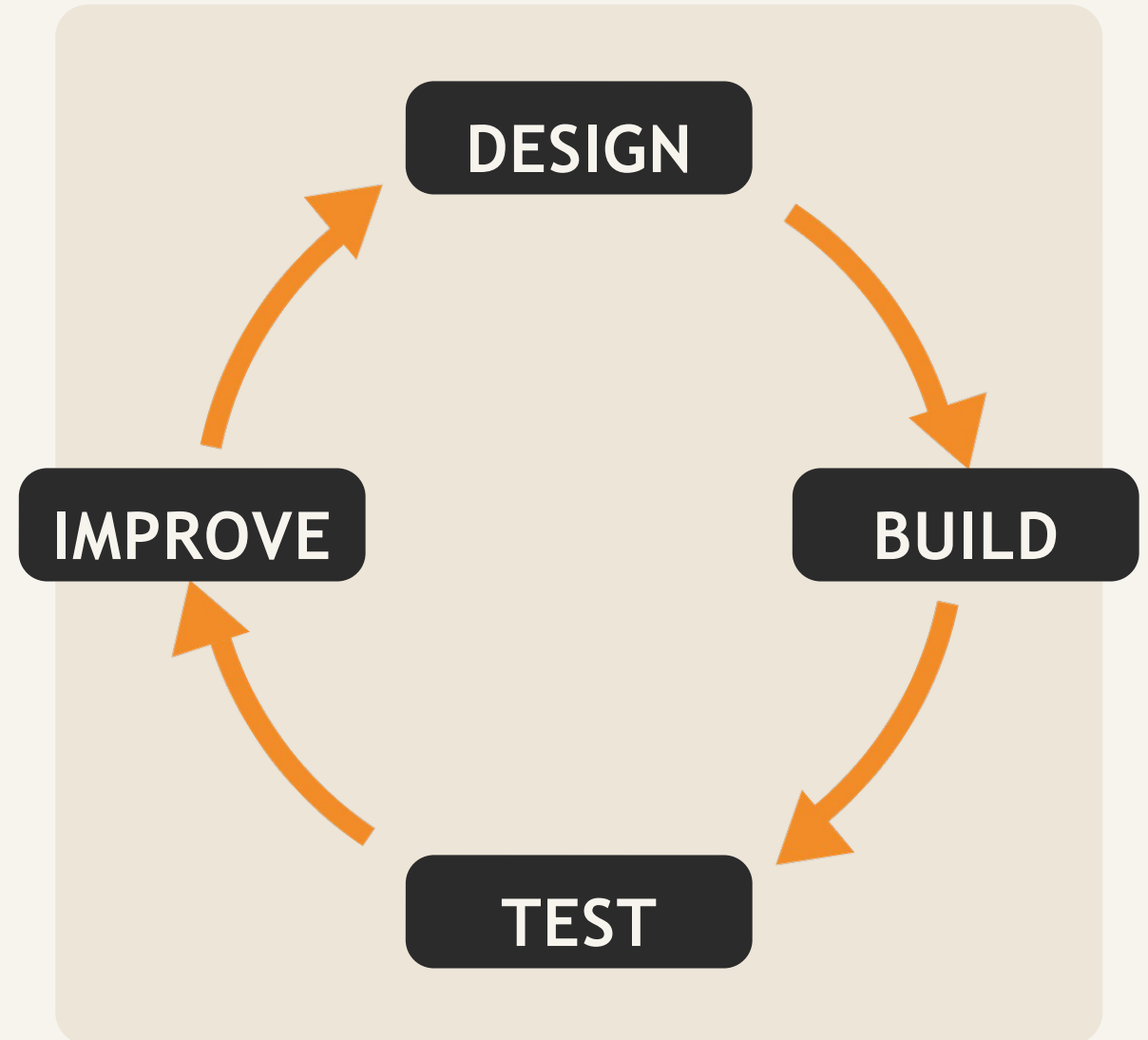
a first working version, built to be tested

Test

try it and measure it to find the faults

Improve

change it, then test again



Worked Example – Your Marble Run

Design

sketch the run on the board

Build

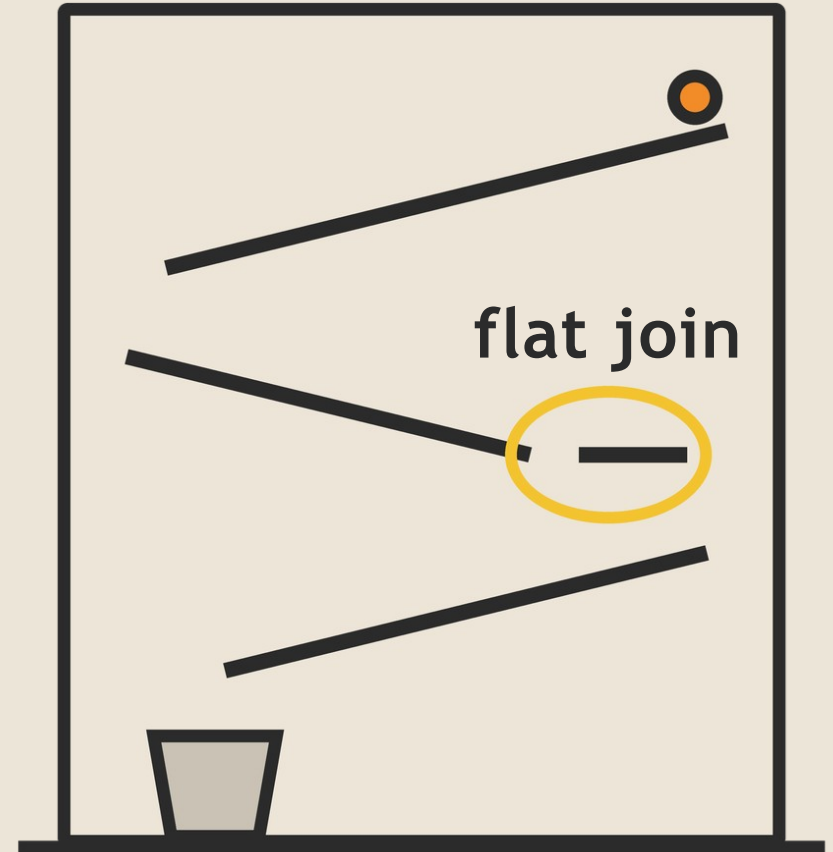
build the top section first

Test

the marble stops on a flat join

Improve

gentler join – then test again



Show Me — The Design Cycle



1 What is a prototype?

A first working version,
built to test

2 Put in order: test, improve, design,
build.

Design, build, test,
improve

3 A prototype drone crashes in testing.
Which stage comes next?

Improve

Show Me — The Design Cycle



4 Why test again after improving?

To check the change
fixed the fault

5 One thing you tested on your marble
run — and what you improved.

Your own answer

Show Me — The Design Cycle

EXTRA PRACTICE



1

Which stage? “Sketches three ideas and picks one.”

Design

2

Which stage? “Times three runs of the marble.”

Test

3

Why is the cycle a loop, not a line?

You improve, then test again

Show Me — The Design Cycle

EXTRA PRACTICE



4

A new school chair wobbles in testing. Improve it — what + why.

Add a brace — **so it stays steady**

5

Why build one prototype before making 1000?

To find faults before they cost a lot

What do engineers do?



- A fix broken machines
- B design, build, test and improve things
- C build things someone else has designed
- D do experiments in a lab

Answer B: Engineers design, build, test and improve – a mechanic repairs.

A company needs a coating that stops a bridge rusting.
Which engineer?



- A civil
- B chemical
- C mechanical
- D electrical

Answer B: The task decides the engineer: a coating is a material – chemical.

Who designs the circuit that controls a lift's motor?



- A mechanical
- B civil
- C electrical
- D chemical

Answer C: It moves, but a circuit is electrical. The motor's gears are mechanical.

A prototype fails its test. What happens next?



- A the project is scrapped
- B it is sold anyway and fixed later
- C a new team builds it again the same way
- D the design is improved, then tested again

Answer D: A failed test isn't the end — it tells you what to improve.

“Describe one task a civil engineer would carry out on a new school.” Which answer gets the mark?



- A “builds it”
- B “civil engineering”
- C “works on the school”
- D “designs the foundations so it won't sink”

Answer D: Only D says what the task is AND why it matters.

Mini Test 1



Mini Test 1 sheet

On your own, in silence, pen only.

8 minutes – 6 marks.

Then swap and mark from the next slides.

Mini Test 1 – Marking

ANSWERS

Q1 (2 marks)

Electrical, mechanical, civil, chemical. All four = 2 marks. Two or three = 1 mark.

Q2 (1 + 1 marks)

(a) Electrical: designs the lighting, the wiring or the power for the pool pumps.

(b) Chemical: designs the water treatment that keeps the pool clean.

Mini Test 1 – Marking

ANSWERS

Q3(a) (1 mark)

Design → build → test → improve

Q3(b) (1 mark)

Change the design or material, like thicker corners or tougher plastic. Then test it again to check it no longer cracks.

Scores 0

“improve it” on its own – say what you change and why. A task given to the wrong engineer scores 0 too.

Brain Dump



In your jotter

Write everything you remember about engineers and what they do.

On your own, 2 minutes. No notes.

Then we check it against the Plenary.

Plenary – Name, Describe, Explain



1

Name the four main types of engineering.

Electrical, mechanical,
civil, chemical

2

One task a chemical engineer does.

Develops fuels, plastics
or coatings

3

The four stages, in order.

Design, build, test,
improve

Plenary – Name, Describe, Explain



4

What + why: one job a civil engineer does.

Designs a bridge – **so it carries traffic safely**

5

What is a prototype for?

Testing, to find the faults

Plenary – Name, Describe, Explain

EXTRA PRACTICE



- 1 Which engineer designs a car's engine?
- 2 Which engineer designs a dam?
- 3 Which stage finds the faults?

Mechanical

Civil

Test

Plenary – Name, Describe, Explain

EXTRA PRACTICE



4 Improve: “engineers make things”.

They design, build, test
and improve

5 Which engineer designs the wiring in
a house?

Electrical

Look back — by the end of this part...



Rate each one 1-3 again. Compare it with your rating at the start.

- 1 I can name the 4 main types of engineering
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LOOKING AHEAD

Lesson 2 — Impacts of Engineering

Good and bad effects on people, money and the environment.

Credits

The big question (Edinburgh Trams): © artful,
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**One Project, Many Engineers — Edinburgh
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